

JOE (JIAZHOU) LIANG



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SUMMARY AND SKILL

- Experienced data scientist with academic research and industry experience, specializing in deep learning, natural language processing (NLP), information retrieval, temporal clustering, and recommender systems.
- Proficient in multiple programming languages including Python and SQL, with popular data analytics (Pandas, Numpy, and Matplotlib) and machine learning (Scikit-learn, Torch) libraries.
- Solid foundation in statistical analysis, with experience in analyzing complex datasets across diverse domains, such as education, census, climate, unstructured text, and various other fields.
- Outstanding communication and presentation skills with proven interpersonal skills in previous jobs and academia.

EDUCATION

Master of Engineering

University of Toronto

09/2023 - Present

- Research Focus in Applied Machine Learning and Data Analytics.
- Completed Research project in D3M Lab, supervised by Professor Scott Sanner.

Bachelor of Mathematics

University of Waterloo

09/2018 - 06/2022

- Dual Programs in Statistics and Computational Math.
- Graduation with Distinction.
- University of Waterloo Merit Scholarship.

WORK EXPERIENCE

Research Assistant | University of Toronto

09/2023 - Present

- Perform quantitative analysis using Python to enhance more than 100 students' experience and outcomes using data from University of Toronto Biology Courses.
- Create visually informative reports with Matplotlib and Plotly to encapsulate outcomes, emphasizing the analysis of student learning habits, and the correlation between grades and the utilization of materials from 10 weeks of lectures through the semester.
- Streamline study replication for future researchers by incorporating Python objective-oriented programming to create analytics embedding class.

Associate Software Engineer | Infosys Canada

06/2022 - 08/2023

- Developed backend service layers encompassing business and service layers, delivering tailored solutions for clients utilizing C# and the .NET framework.
- Designed scalable query applications incorporating data extraction from Microsoft SQL Server and utilizing Entity Framework.
- Maintained ongoing client communication, collecting feedback and addressing concerns across industries.

Data Analyst | T School Education

06/2018 - 09/2018

- Developed an automated data processing framework using Excel to assess information from raw datasets for over 3000 primary school students.
- Optimized daily lecture preparation workflow, achieving a 30% reduction in time by utilizing historical data and conducting company-wide surveys.

PROJECTS

A Generalized Framework for Optimal Temporal Clustering

09/2023 - Present

- Conduct research on the MILP optimization of clustering algorithms for time-series data in the University of Toronto D3M Lab, supervised by Prof Scott Sanner.
- Introduce six types of algorithms that demonstrate superior performance compared to widely used clustering methods through synthetic data experiments, effectively minimizing the maximum distance between data points and cluster centers while preserving high data purity.
- Perform experiments with proposed algorithms in practical settings, such as climate data, census data, and business establishment records. These experiments yielded valuable insights with potential applications in relevant fields.

Unfair ToS: A GPT Based Unfair Contract Terms Detector

09/2023 - Present

- Design and implement a GPT-based framework to identify critical and ambiguous terms within Terms of Service (ToS) at the sentence level.
- Attain a high F1-score in the classification of 'unfair' terms through fine-tuning the GPT-2 model based on 100 annotated Tos, outperforming the BERT baseline model.
- Enhance the user experience by utilizing GPT-4 to automatically highlight essential terms in raw ToS, creating a summary to increase readability, leading to the creation of more user-friendly and easily understood versions.

Classification on Financial Health Survey

01/2022 - 05/2022

- Classified samples based on Germany General Survey 2016 with 2000 features into 5 categories using machine learning models. Such as but not limited to neural networks, SVM, and decision tree-based models (random forest).
- Improved cross-entropy score by 10% by applying dimension reduction techniques, such as principal component analysis. Implemented stack generalization to multiple skillful individual learners.